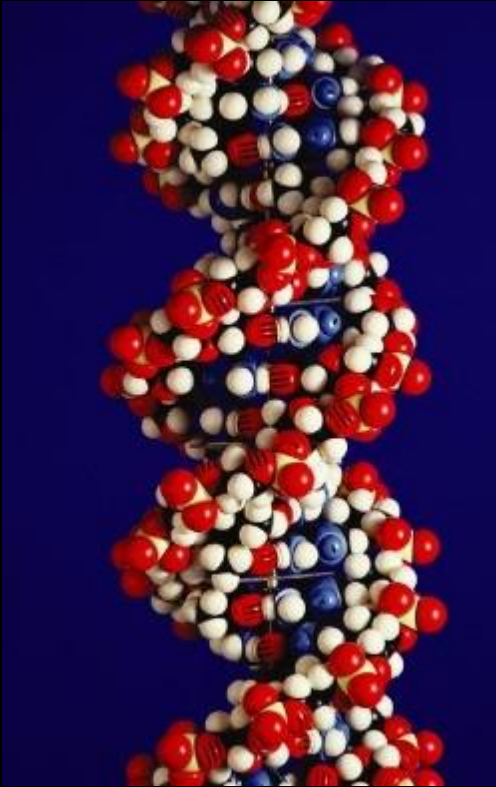


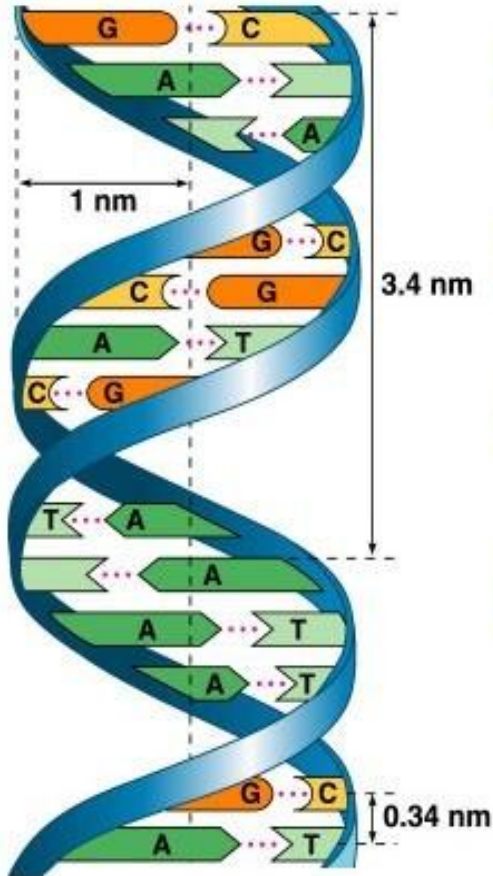
DNA Replication



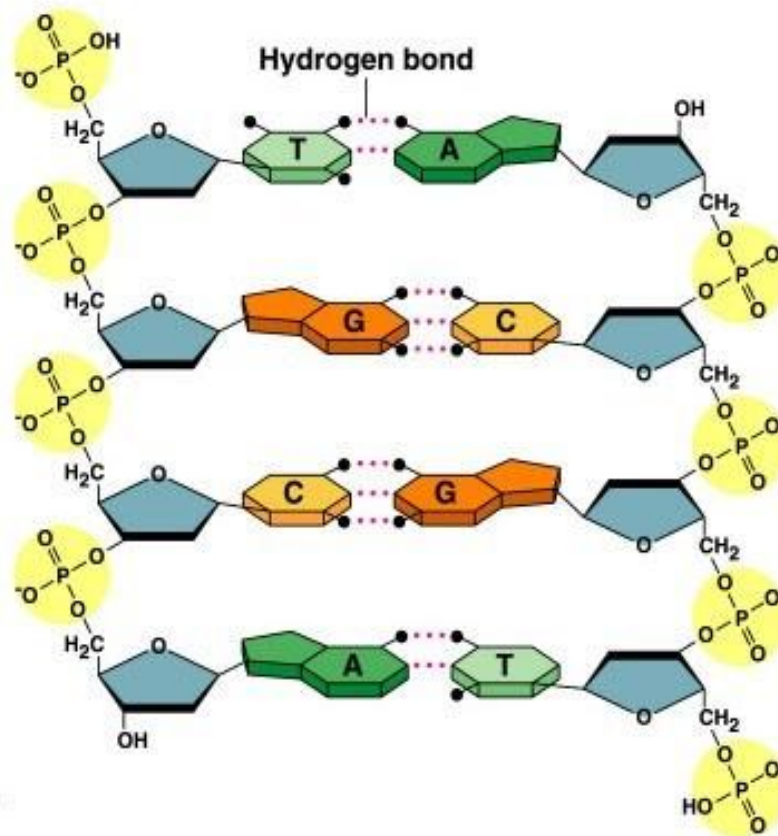
DNA – DOUBLE HELICAL STRUCTURE WATSON and CRICK Model



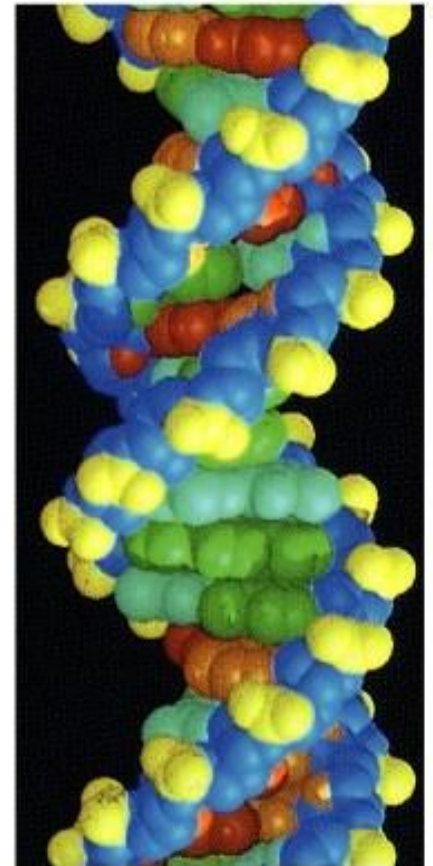
DNA – DOUBLE HELICAL STRUCTURE



(a) Key features of DNA structure

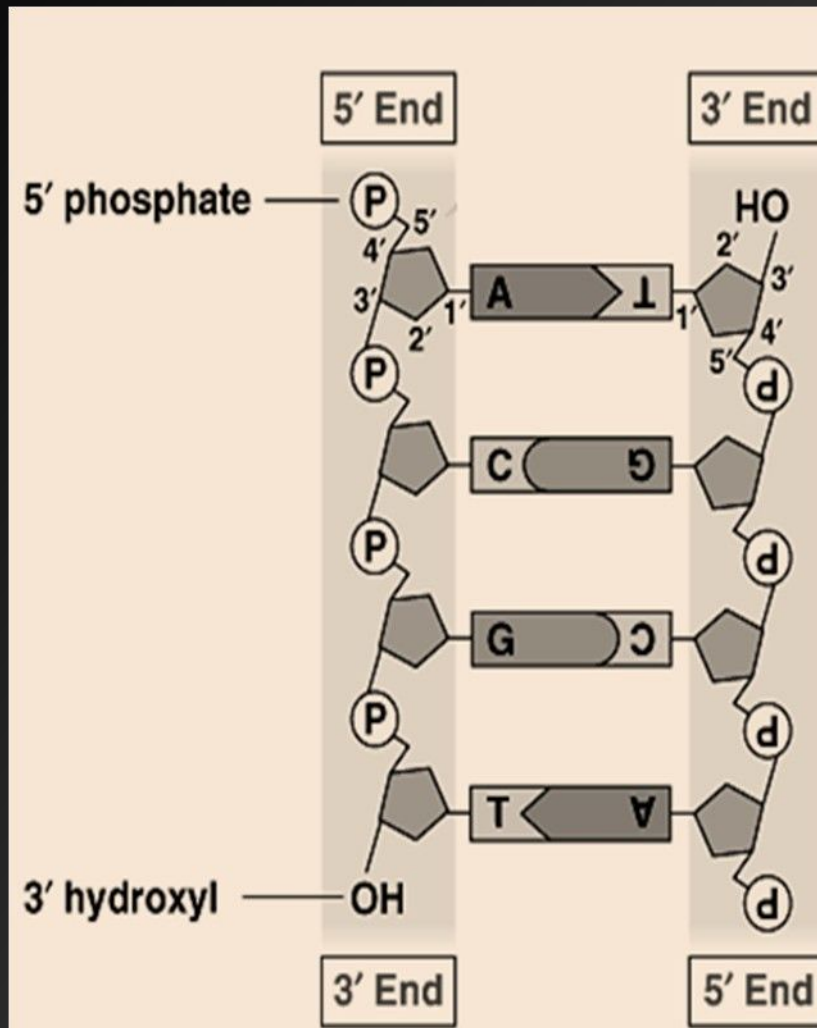


(b) Partial chemical structure



(c) Space-filling model

Directionality of DNA



Nucleotides in DNA backbone are bonded together by phosphodiester linkage between 3' & 5' carbons.

DNA molecule has “direction.”

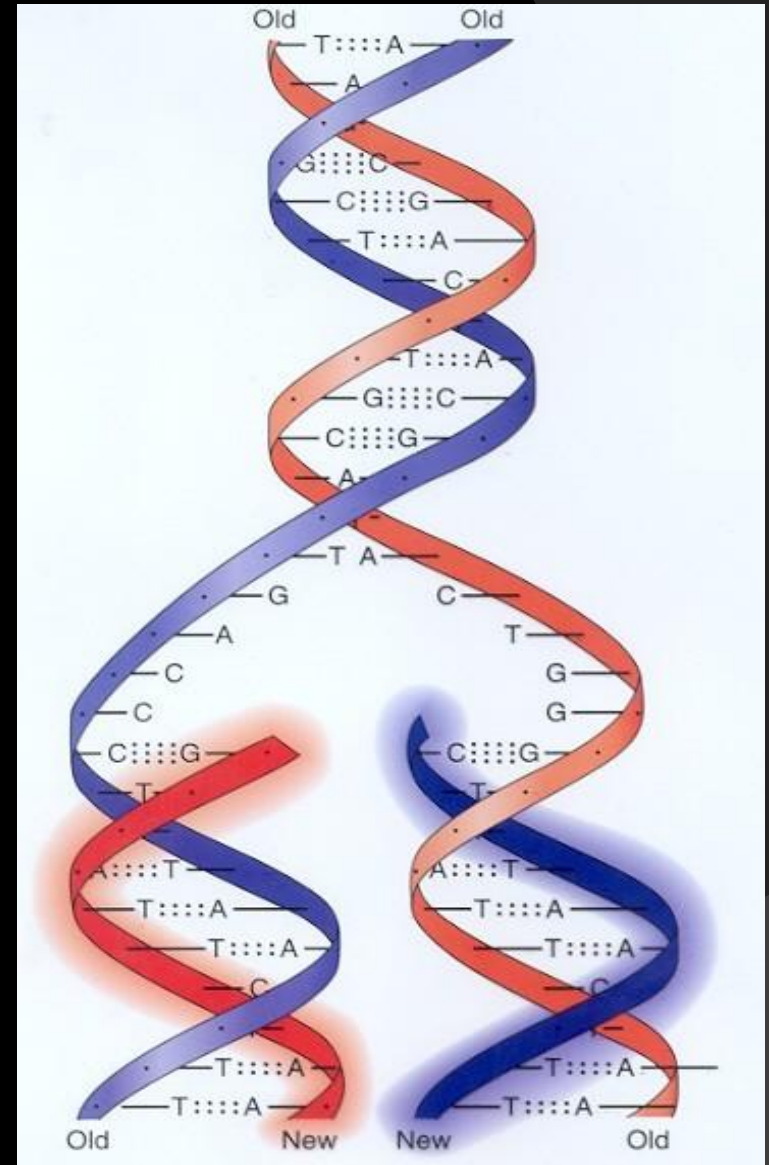
Complementary strands run in opposite directions.

DNA Replication - Introduction

- Basis for inheritance
- Fundamental process occurring in all cells for copying DNA to transfer the genetic information to daughter cells
- Each cell must replicate its DNA before division.

DNA Replication

- Semi conservative
- Parental strands are not degraded
- Base pairing allows each strand to serve as a template for a new strand
- New duplex is 1/2 parent template & 1/2 new DNA



DNA Replication

- Semi discontinuous
- Leading & Lagging strands

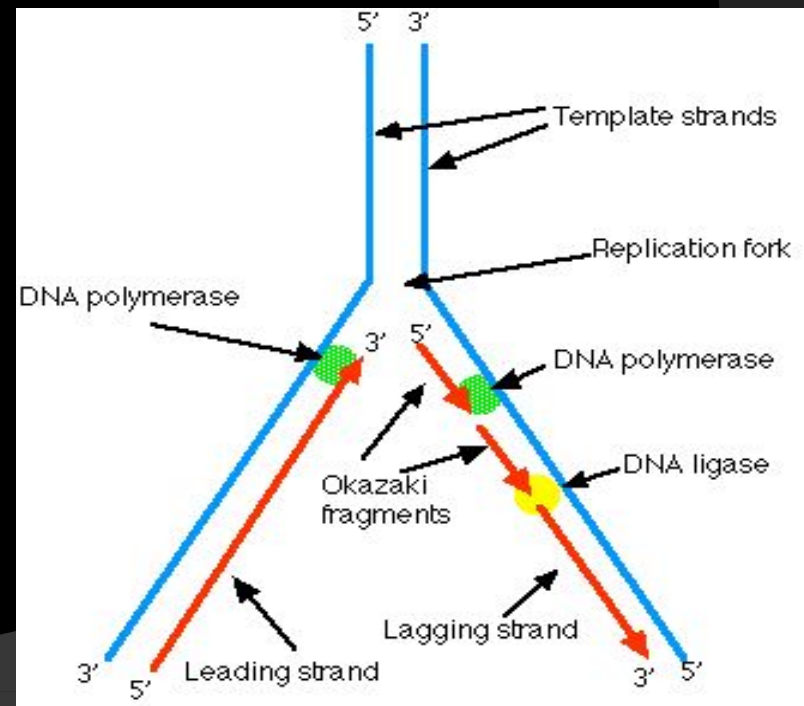
Leading strand

- continuous synthesis

Lagging strand

- Okazaki fragments
- joined by ligases

Okazaki



DNA Replication

- Energy of Replication
- The nucleotides arrive as
- nucleoside triphosphates
 - DNA base, sugar with PPP
 - P-P-P = energy for bonding
 - DNA bases arrive with their own energy source for bonding
 - bonded by enzyme: DNA polymerase III

DNA Replication

- Primer is needed
 - DNA polymerase can only add nucleotides to 3'end of a growing DNA strand
 - need a “starter” nucleotide to make a bond
- strand only grows 5 “ → 3 “
- Template is read in the 3'-5' direction while polymerization takes place in the 5' → 3' direction

Primer

RNA primer

- Synthesized by **Primase**
- serves as a starter sequence for DNA polymerase III
- Only one RNA Primer-required for the leading strand
- RNA Primers for the lagging strand depend on the number of **“OKAZAKI FRAGMENTS”**
- RNA Primer has a free 3'OH group to which the first Nucleotide is bound.

DNA Replication - Steps

- Identification of the origins of replication
- Unwinding (denaturation) of dsDNA to provide an ssDNA template
- Formation of the replication fork
- Initiation of DNA synthesis and elongation
- Primer removal and ligation of the newly synthesized DNA segments
- Reconstitution of chromatin structure

Components of Replication

- **DNA polymerases** - Deoxynucleotide polymerization
- **Helicase** - Processive unwinding of DNA
- **Topoisomerases** - Relieve torsional strain that results from helicase-induced unwinding
- **RNA primase** - Initiates synthesis of RNA primers
- **Single-strand binding proteins** - Prevent premature reannealing of dsDNA
- **DNA ligase** - Seals the single strand nick between the nascent chain and Okazaki fragments on lagging strand

Origin of Replication - Prokaryotes

- At the **origin of replication (ori)**, there is an association of sequence-specific dsDNA-binding proteins with a series of direct repeat DNA sequences.
- In *E coli*, the oriC is bound by the **protein dnaA**.
- A complex is formed consisting of 150–250 bp of DNA and multimers of the DNA-binding protein. This leads to the local **denaturation and unwinding of an adjacent A+T-rich region of DNA**.

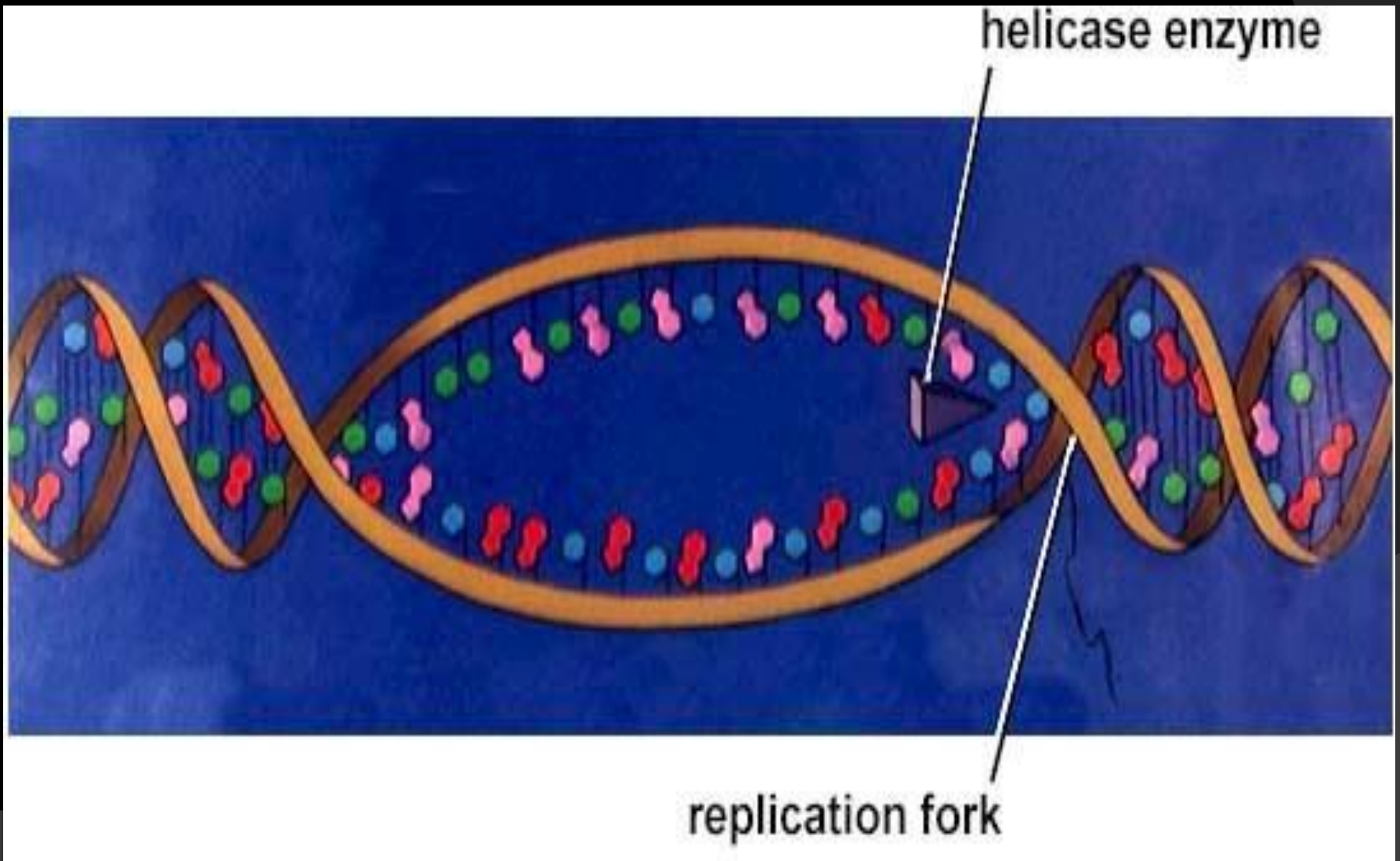
Origin of Replication - Eukaryotes

- Functionally similar **autonomously replicating sequences (ARS) or replicators** have been identified in yeast cells.
- The ARS contains a somewhat degenerate 11-bp sequence called the **origin replication element (ORE)**.
- The ORE binds a set of proteins, analogous to the dnaA protein of *E coli*, which is collectively called the **origin recognition complex (ORC)**.
- The ORE is located adjacent to an approximately 80-bp A+T-rich sequence that is easy to unwind. This is called the **DNA unwinding element (DUE)**.

Unwinding of DNA

- The interaction of proteins with *ori* defines the start site of replication and provides a short region of ssDNA essential for initiation of synthesis of the nascent DNA strand.
- **DNA Helicase** allows for processive unwinding of DNA.
- **Single-stranded DNA-binding proteins (SSBs)** stabilize this complex.
- In cooperation with SSB, this leads to DNA unwinding and active replication.

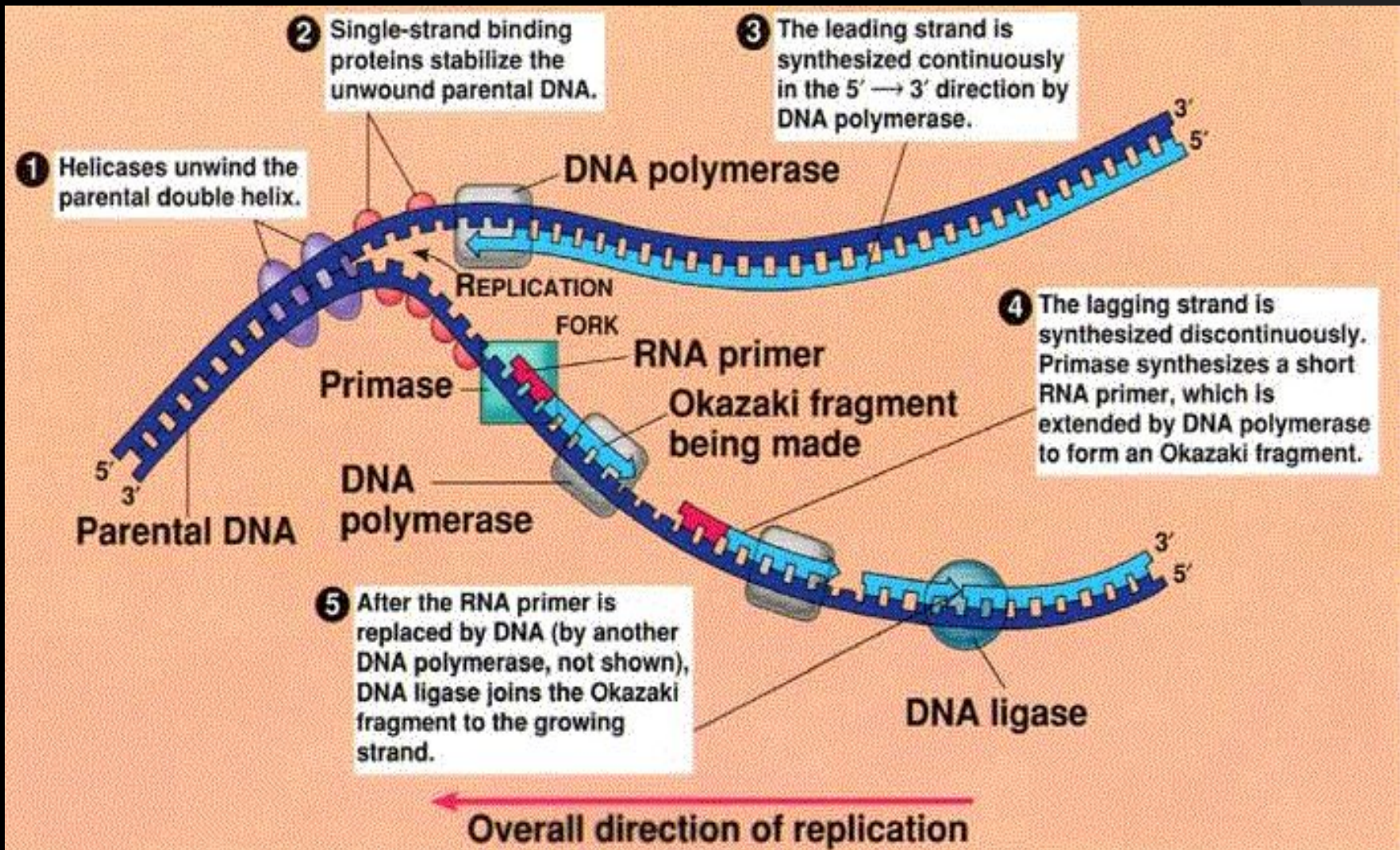
Unwinding of DNA



Formation of the Replication Fork

- The polymerase III holoenzyme binds to template DNA as part of a multiprotein complex
- DNA polymerases only synthesize DNA in the 5' to 3' direction,
- Because the DNA strands are antiparallel , the polymerase functions asymmetrically.
- On the leading (forward) strand, the DNA is synthesized continuously.
- On the lagging (retrograde) strand, the DNA is synthesized in short (1–5 kb) fragments, the so-called Okazaki fragments.

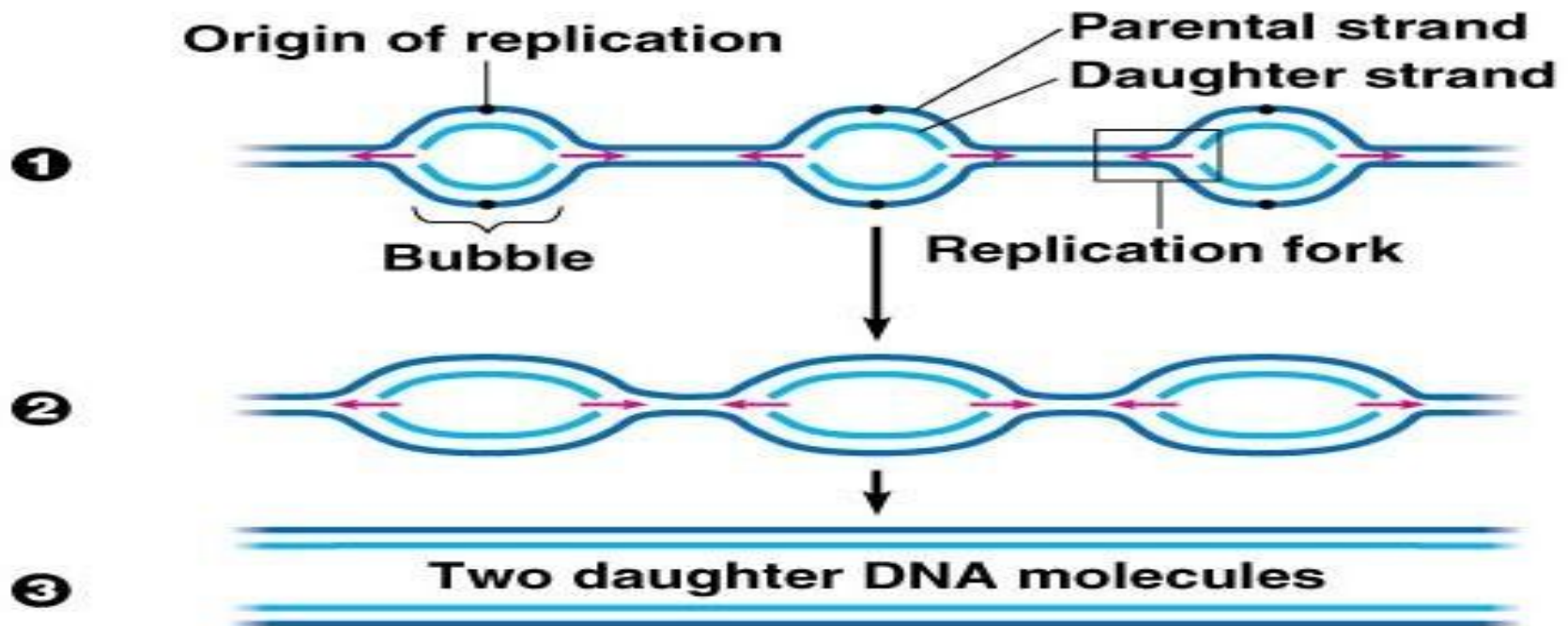
Replication Fork



Formation of Replication Bubbles

- Replication occurs in both directions along the length of DNA and both strands are replicated simultaneously.
- This replication process generates **"replication bubbles"**

Replication Bubbles



(a) In eukaryotes, DNA replication begins at many sites along the giant DNA molecule of each chromosome.

The DNA Polymerase Complex

- A number of different DNA polymerase molecules engage in DNA replication. These share three important properties: (1) **chain elongation**, (2) **Processivity**, and (3) **proofreading**.
- **Chain elongation** accounts for the rate (in nucleotides per second) at which polymerization occurs.
- **Processivity** is an expression of the **number of nucleotides added** to the nascent chain before the polymerase disengages from the template.
- The **proofreading** function identifies **copying errors and corrects** them

DNA Polymerase Complex

- In *E coli*, **polymerase III** (pol III) functions at the replication fork. Of all polymerases, it catalyzes the highest rate of chain elongation and is the **most processive**.
- **Polymerase II** (pol II) is mostly involved in **proofreading and DNA repair**.
- **Polymerase I** (pol I) **completes chain synthesis between Okazaki fragments** on the lagging strand.

Differences between DNA Polymerase I, II and III

	DNA pol I	DNA pol II	DNA pol III
Polymerization Rate	Low	Low	High
Processivity	Low	Low	High
Proof reading	3'-5' and 5'-3' Exonuclease activities	3'-5' Exonuclease activity	3'-5' Exonuclease activity
Primer removal	Best	Nil	Nil
Strand synthesis	Lagging strand	No role	Both strands
DNA repair	Active	Active	No role

Eukaryotic DNA polymerases

- Eukaryotic cells have counterparts for each of these enzymes plus some additional ones. A comparison is shown

<i>E coli</i>	Mammalian	Function
I	Alpha	Gap filling and synthesis of lagging strand
II	Epsilon	DNA proofreading and repair
	βeta	DNA repair
	Gamma	Mitochondrial DNA synthesis
III	delta	Processive , Leading strand synthesis

Initiation & Elongation of DNA Synthesis

- **Primer** - The priming process involves the nucleophilic attack by the 3'-hydroxyl group of the RNA primer on the phosphate of the first entering deoxynucleoside triphosphate with the splitting off of pyrophosphate.
- **Mammalian DNA polymerase Alpha** is mainly responsible for the synthesis of primer.

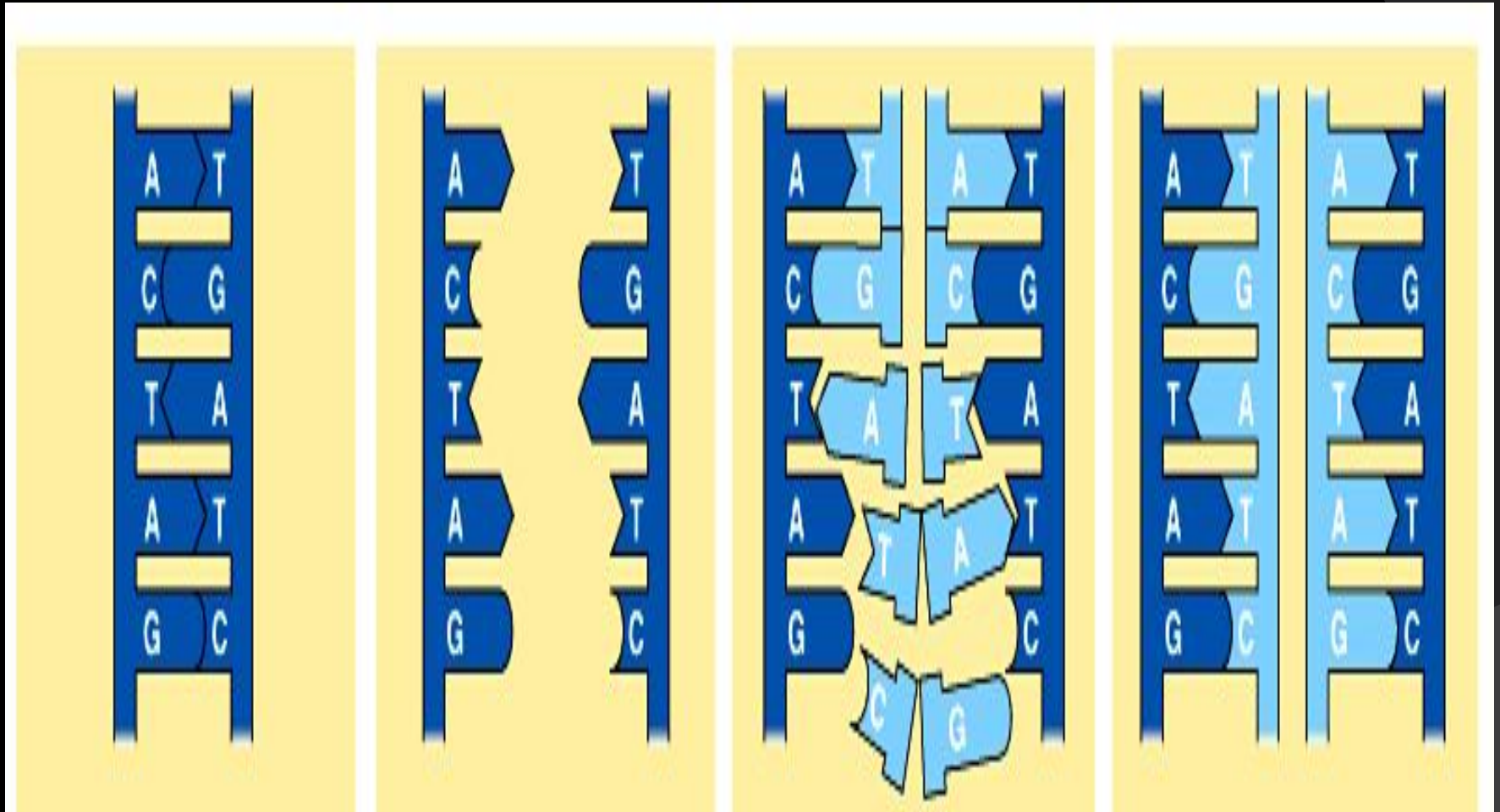
Initiation & Elongation of DNA Synthesis

- Selection of the proper deoxyribonucleotide whose terminal 3'-hydroxyl group is to be attacked is dependent upon **proper base pairing with the other strand** of the DNA molecule according to the rules proposed originally by **Watson and Crick**

Initiation & Elongation of DNA Synthesis

- When an adenine deoxyribonucleoside monophosphoryl moiety is in the template position, a thymidine triphosphate will enter and its phosphate will be attacked by the 3'-hydroxyl group of the deoxyribonucleoside monophosphoryl most recently added to the polymer.
- By this stepwise process, the template dictates which deoxyribonucleoside triphosphate is complementary and by hydrogen bonding holds it in place while the 3'-hydroxyl group of the growing strand attacks and incorporates the new nucleotide into the polymer.

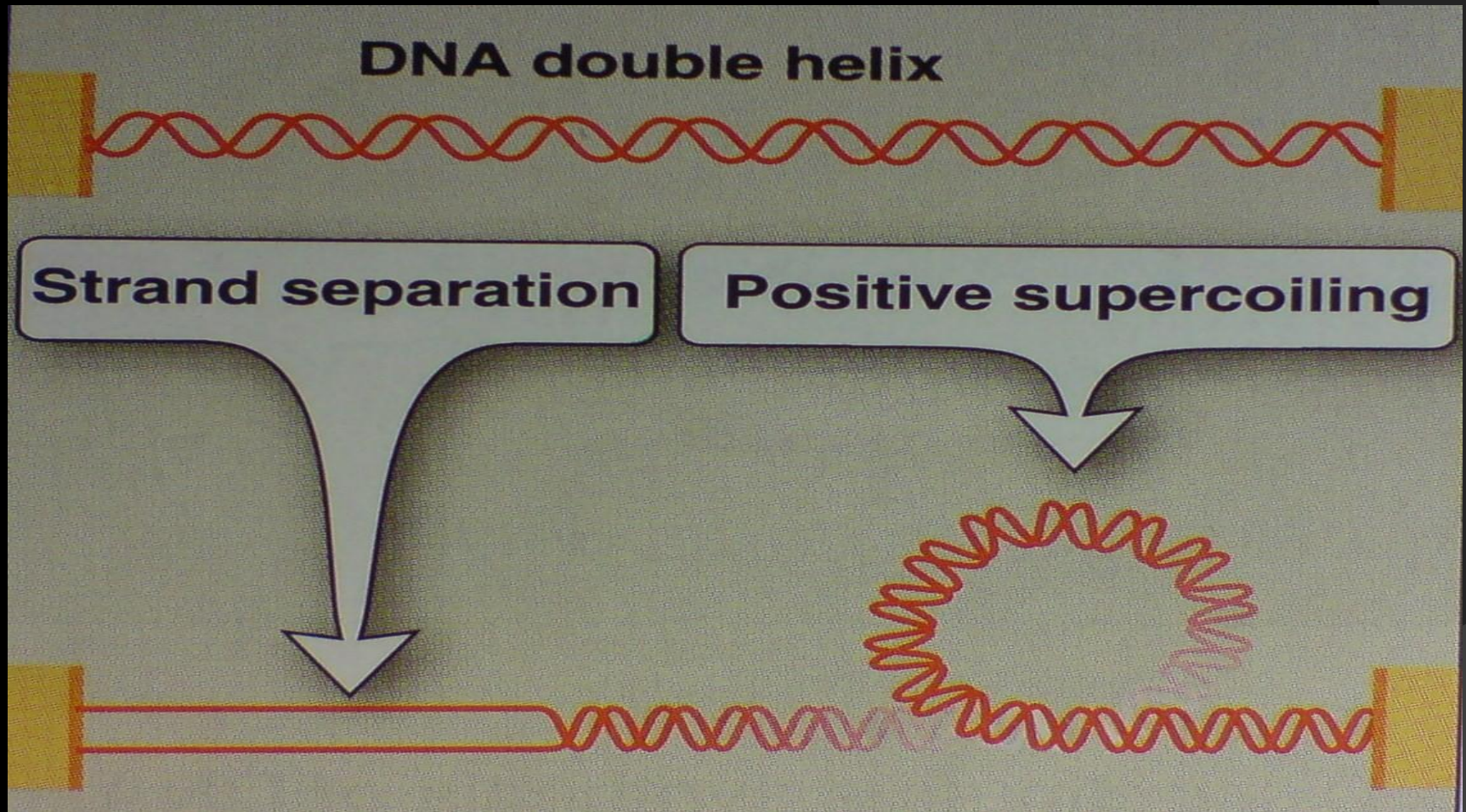
Base pairing in DNA Replication



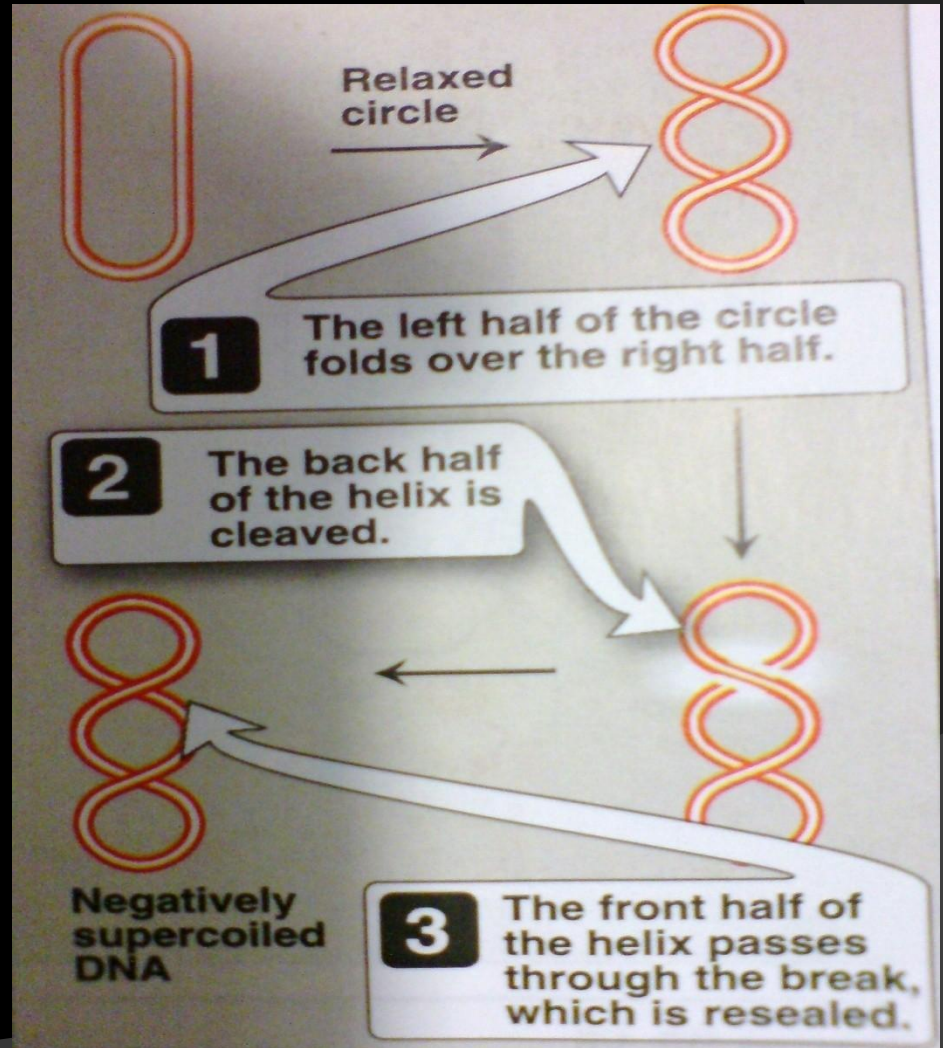
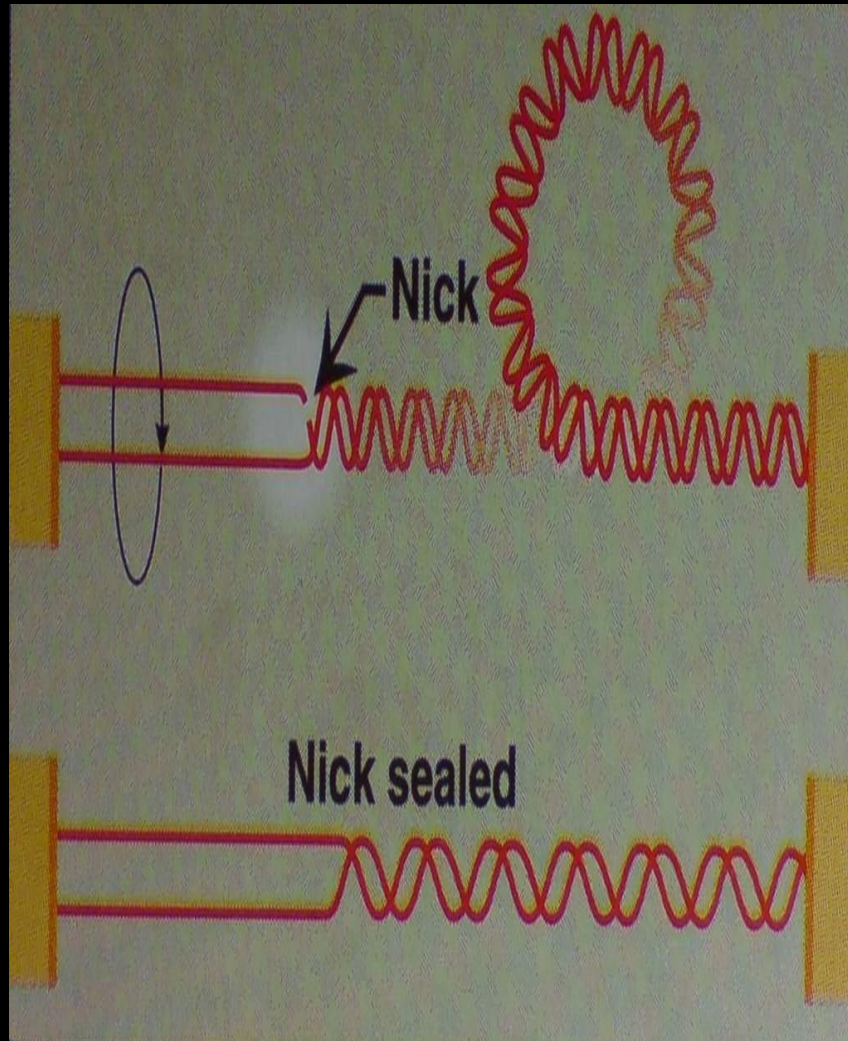
DNA Topoisomerases

- Relief of super coils is done by Topoisomerases
- Two types:
- **Topoisomerases I** : acts by making a transient single cut in the backbone of the DNA, enabling the strands to swivel around each other to remove the build-up of twists
- **Topoisomerase II (DNA Gyrase)** acts by introducing double stranded breaks enabling one double-stranded DNA to pass through another, thereby removing knots and entanglements that can form within and between DNA molecules.

Formation of super coils



Mechanism of action of Topoisomerases (Type-I and Type-II)



Primer removal and Nick sealing

- Primers are removed by DNA polymerase I by replacing ribonucleotides with deoxy Ribonucleotides
- Nicks are sealed by DNA ligase
- Multiple primers on the Lagging strand while single primer on the leading strand.

Proof-reading and Editing

- 1000 bases/second =
lots of typos!
 - DNA polymerase I
 - proofreads & corrects typos
 - repairs mismatched bases
 - removes abnormal bases
 - repairs damage throughout life
- reduces error rate from 1 in 10,000 to
1 in 100 million bases

Termination of replication

- In prokaryotes:

DNA replication terminates when replication forks reach specific “termination sites”.

- the two replication forks meet each other on the opposite end of the parental circular DNA .

Termination of replication

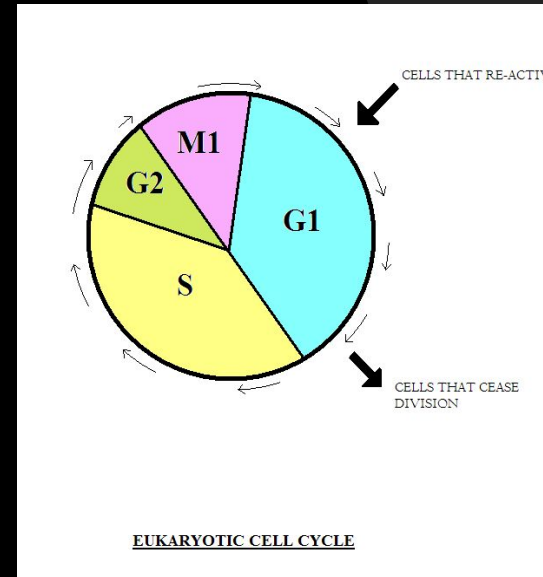
- This process is completed in about 30 minutes, a replication rate of 3×10^5 bp/min in prokaryotes
- The entire mammalian genome replicates in approximately 9 hours, the average period required for formation of a tetraploid genome from a diploid genome in a replicating cell.

Reconstitution of Chromatin Structure

- Chromatin structure must be re-formed after replication. Newly replicated DNA is rapidly assembled into **nucleosomes**, and the preexisting and newly assembled **histone octamers** are randomly distributed to each arm of the replication fork.

DNA Synthesis and the Cell Cycle

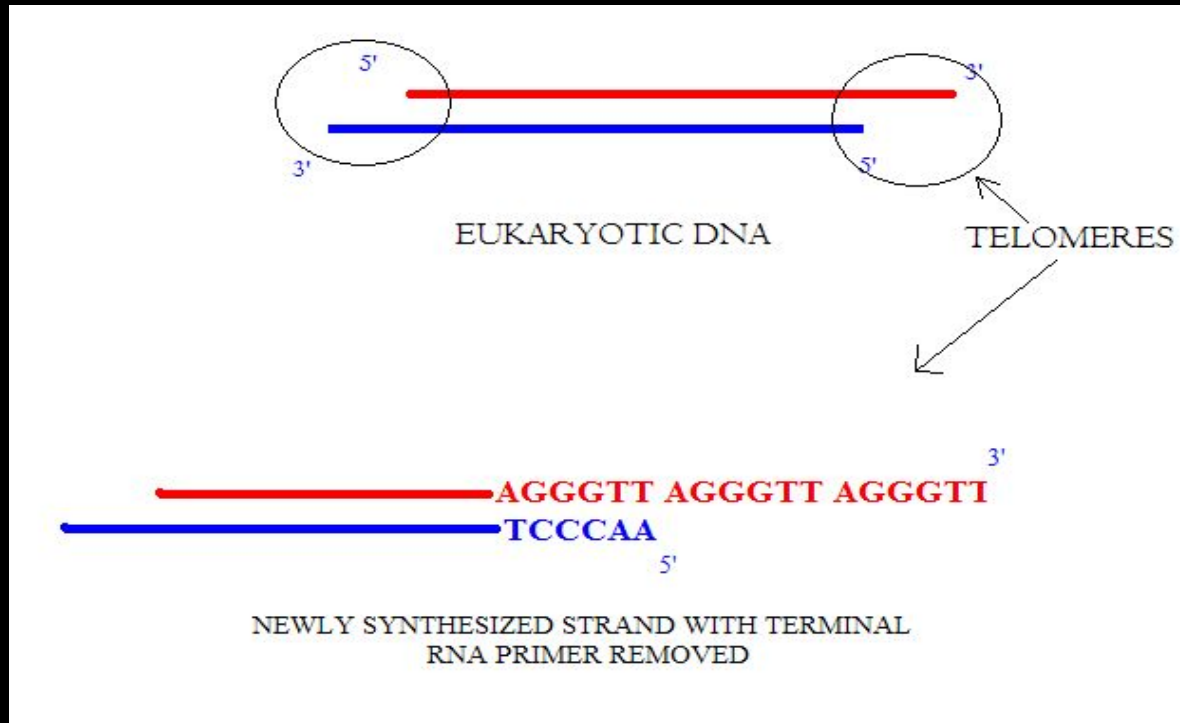
- In animal cells, including human cells, the replication of the DNA genome occurs only during the **synthetic or S phase**.
- This is usually temporally separated from the mitotic phase by non synthetic periods referred to as **gap 1 (G1)** and **gap 2 (G2)**, occurring before and after the S phase, respectively
- The cell prepares for DNA synthesis in G1 and for mitosis in G2.



Telomeres

- In eukaryotic replication, following removal of RNA Primer from the 5' end of lagging strand; a gap is left.
- This gap exposes DNA strand to attack of 5' exonucleases.
- This problem is overcome by **Telomerase.**

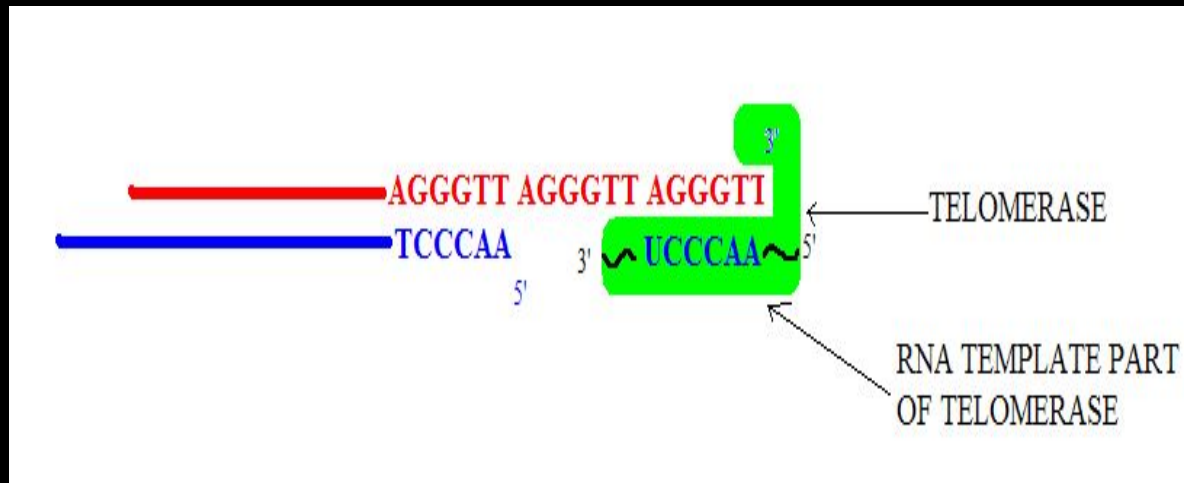
Telomeres



Repeating, non-coding sequences at the end of chromosomes = protective cap

- limit to ~50 cell divisions

Mechanism of action of Telomerase



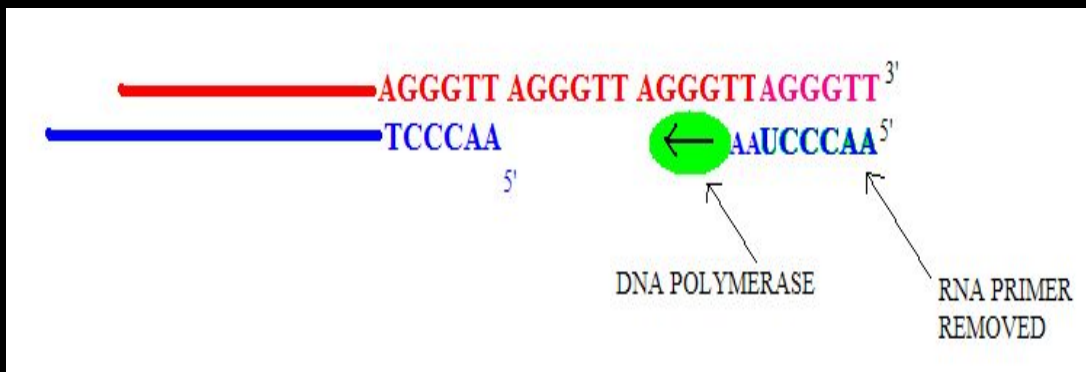
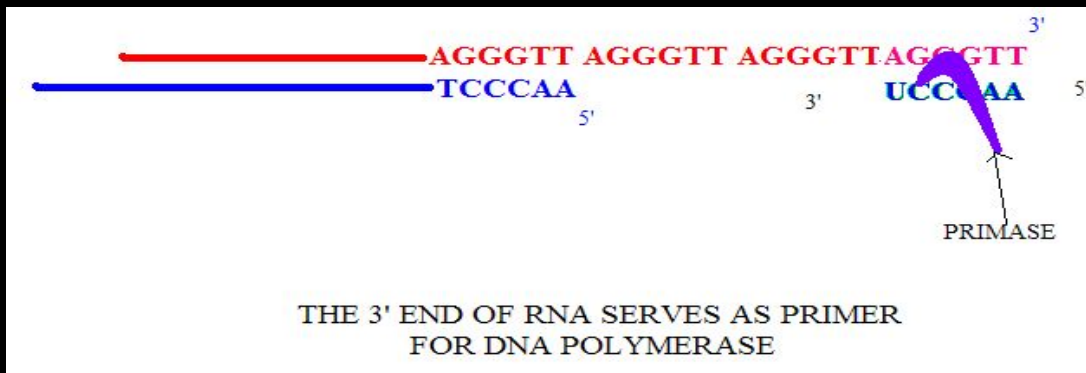
The enzyme synthesizes (TTAGGG) n repeats on to the Telomere sequences, using an internal RNA template.

Mechanism of action of Telomerase (Contd.)



Telomerase acts like a reverse Transcriptase. It recognizes 3' end of telomere, based on the RNA component, a small DNA strand is synthesized

Mechanism of action of Telomerase (Contd.)



Telomerase

- different level of activity in different cells
 - High in stem cells & cancers
 - Activity lost in old age
 - Potential target for newer anticancer drugs

Inhibitors of DNA replication

- Bacterial DNA Gyrase(Type II Topoisomerase) - Inhibited by **Novobiocin and Nalidixic acid**.
- **Ciprofloxacin** interferes with DNA breakage and rejoining process
- Mammalian topoisomerases – inhibited by **Etoposide and Adriamycin**, used as anticancer drugs.
- Nucleoside analogues also inhibit replication and are used as anticancer drugs.

Summary of Replication

- **Unwinding**- forms replication fork
- **Primase**- Synthesizes RNA primer
- **Continuous synthesis** -Leading strand
- **Discontinuous synthesis** – Lagging strand (Okazaki fragments)
- **Synthesis 5'-3'** direction
- **Primers** removed, nick sealed
- **Proof reading** by DNA polymerases
- **organized into chromatin** structure